



AI-BASED BUSINESS DECISION SUPPORT SYSTEM

Data-Driven Insights for Smarter Business Decisions



OBJECTIVE

To develop an AI-powered Business Decision Support System using Power BI and ChatGPT to analyze e-commerce sales data and provide actionable insights that help improve performance, optimize operations, and support strategic decision-making.

TOOLS USED

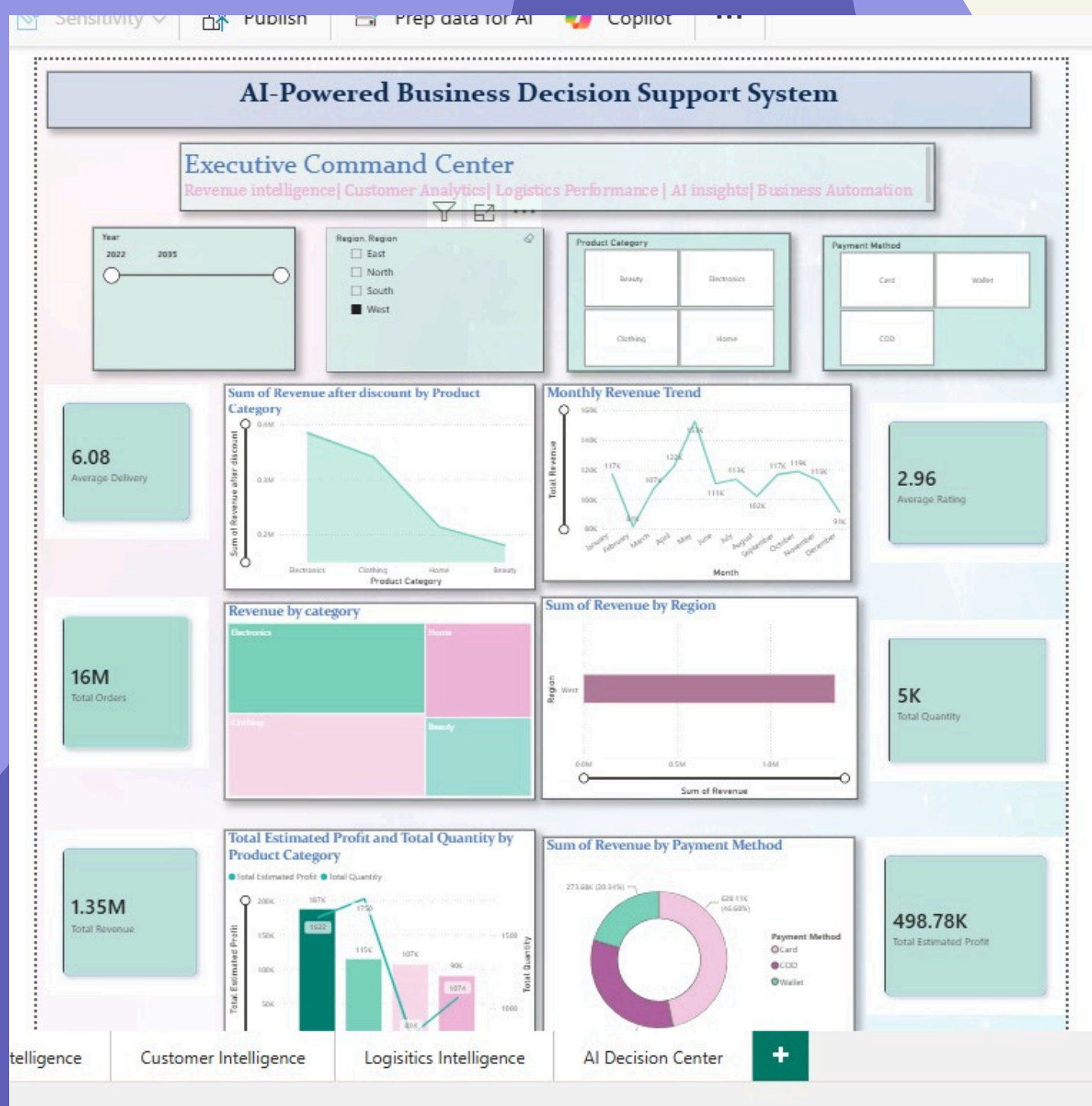
- Excel
- Power BI
- ChatGPT
- Canva
- Zapier
- GitHub

DATASET

E-commerce Sales Analytics
Dataset (5000 Records)

Features Include:

- Order ID
- Product Category
- Region
- Revenue
- Customer Rating
- Payment Method
- Delivery Days
- Discounts



AI-GENERATED BUSINESS INSIGHTS

- Revenue shows a consistent growth trend with peak sales in July.
- West region is the top contributor, followed by North and South.
- Electronics and Fashion are the most profitable categories.
- Average customer rating of 4.21 indicates overall good satisfaction.
- Faster delivery (4.76 days) positively impacts customer experience.
- Customers prefer digital payments, especially Credit Cards and UPI.
- Moderate discounts (0-10%) are more frequent and effective.
- Data-driven strategies can improve sales, retention and profitability.

STRATEGIC RECOMMENDATIONS

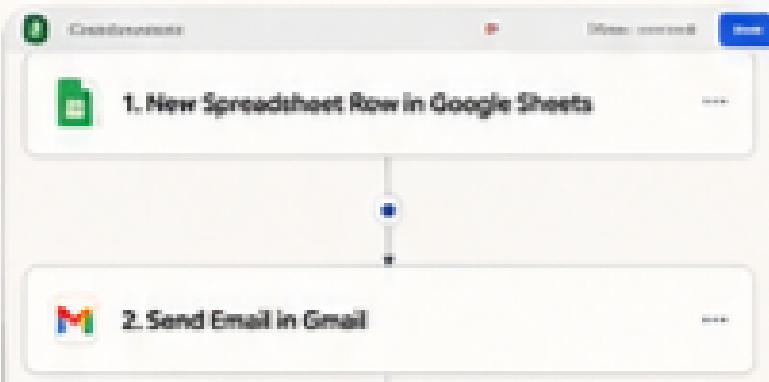
- ✓ Increase marketing efforts in low-performing regions.
- ✓ Focus inventory on high-demand product categories.
- ✓ Optimize delivery operations to reduce delivery time.
- ✓ Run targeted discounts and seasonal promotions.
- ✓ Encourage digital payments through offers & cashback.
- ✓ Improve customer service to increase satisfaction.
- ✓ Use predictive analytics for demand forecasting.
- ✓ Launch loyalty programs to improve customer retention.

AUTOMATION WITH ZAPIER

Workflow: Notify via Email when a new sales record is added in Google Sheets.



- ✓ Saves Time
- ✓ Reduces Manual Work
- ✓ Instant Notifications
- ✓ Improves Efficiency



BUSINESS BENEFITS

- Better Decision-Making
- Improved Operational Efficiency
- Increased Revenue & Profitability
- Enhanced Customer Experience
- Data-Driven Strategic Planning
- Real-Time Business Monitoring

CONCLUSION

This AI-Based Business Decision Support System leverages the power of data visualization, and AI to turn raw data into meaningful insights. It empowers businesses to make faster, smarter, more informed decisions that drive growth and maintain a competitive advantage in the market.

